

Model Testing & Refinement Report

AnalystLab Africa Experience Lab | Week 7 | Data Science

Week 7 result The Week 6 Gradient Boosting candidate was systematically tested, refined and re-tested. Held-out ROC-AUC improved from 0.68 to 0.72.

1. Week 6 to Week 7

Week 6 selected Gradient Boosting as the candidate model. It achieved about 65% accuracy, 65% no-show recall, 65% F1-score and 0.68 ROC-AUC.

Week 7 tested whether this performance was stable, looked for weaknesses, refined the model and checked the result again.

2. Testing approach

- 5-fold Stratified Group K-Fold validation to keep the same patient out of both training and validation folds.
- Cross-validation versus held-out test comparison to check stability.
- Training versus test comparison to check overfitting.
- Hyperparameter tuning of the existing Gradient Boosting model.
- Segment testing by previous no-show history, reminder status, appointment type and booking lead time.
- False-positive and false-negative analysis, followed by a targeted threshold test.

3. Candidate validation

Metric	Mean 5-fold CV	Held-out test
Accuracy	0.62	0.65
Precision	0.63	0.64
No-show recall	0.61	0.65
F1-score	0.62	0.65
ROC-AUC	0.67	0.68

The held-out results were close to the average cross-validation results. This supported reasonable generalization stability. Recall varied more than AUC, so no-show detection still needed closer testing.

4. Overfitting test and refinement

Metric	Original train	Original test	Tuned train	Tuned test
Accuracy	0.69	0.65	0.63	0.66
No-show recall	0.68	0.65	0.66	0.67
F1-score	0.69	0.65	0.65	0.67
ROC-AUC	0.77	0.68	0.70	0.72

The original model showed a 0.09 train-test AUC gap. This suggested moderate overfitting. The model was tuned to reduce unnecessary complexity.

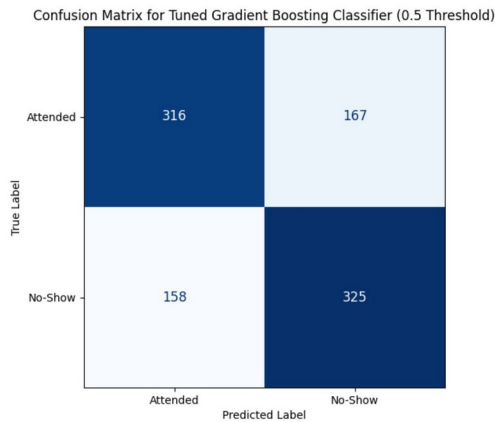
The selected tuned settings were: learning_rate 0.05, max_depth 3, n_estimators 50, subsample 0.9 and max_features None.

Retest: PASS After tuning, held-out AUC increased to 0.72 and no-show recall/F1 increased to 0.67. The earlier train-test AUC gap also disappeared.

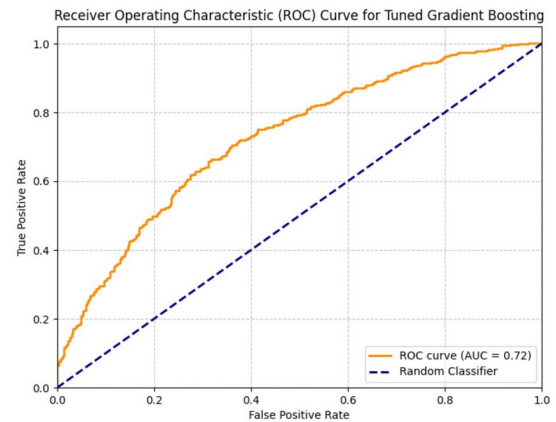
5. Final tuned model

Metric	Week 6 GB	Week 7 tuned GB	Change
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Accuracy	0.65	0.66	+0.01
Precision	0.64	0.66	+0.02
No-show recall	0.65	0.67	+0.02
F1-score	0.65	0.67	+0.02
ROC-AUC	0.68	0.72	+0.04



Confusion matrix at 0.50 threshold



ROC curve - AUC 0.72

At the default threshold, the tuned model correctly classified 316 attended appointments and 325 no-shows. It produced 167 false positives and 158 false negatives.

6. Segment testing

Segment	N	Recall	F1	AUC
No previous no-show	587	0.581	0.609	0.717
Previous no-show	379	0.784	0.729	0.715
Reminder: No	257	0.699	0.706	0.754
Reminder: Yes	709	0.663	0.652	0.712
Diagnostic Test	109	0.800	0.714	0.761
Follow-up	271	0.699	0.688	0.743
General Consultation	406	0.690	0.685	0.737
Specialist Consultation	180	0.532	0.562	0.655

The model performed better for patients with previous no-shows. Specialist Consultations were the weakest appointment-type segment, with 53.2% recall.

7. Booking lead-time weakness

Booking lead	N	Recall @0.50	F1 @0.50	Recall @0.45	F1 @0.45
<=30 days	503	0.230	0.339	0.372	0.444
>30 days	463	0.943	0.779	1.000	0.786

The largest weakness was short-lead appointments. At threshold 0.50, the model detected only 23.0% of no-shows booked within 30 days. Lowering the threshold to 0.45 increased recall to 37.2%, but precision fell from 64.6% to 55.3%.

The lower threshold improved detection, but it did not solve the weakness. The 0.50 threshold remains the default candidate, while 0.45 is documented as a higher-recall alternative.

8. Final error analysis

Prediction type	Mean booking lead	Previous no-show rate
False Negative	17.5 days	0.120
True Negative	14.7 days	0.112

False Positive	41.8 days	0.214
True Positive	43.9 days	0.206

Missed no-shows looked similar to attendees: they had shorter booking lead times and less previous no-show history. The model was stronger when risk patterns were more obvious.

9. Cross-track validation

A Data Analytics collaborator independently checked booking lead time against the underlying appointment outcomes.

Booking lead	Actual no-show rate	Model recall @0.50
<=30 days	36.84%	23.0%
>30 days	60.31%	94.3%

The Analytics result confirmed that longer booking lead time is genuinely associated with a higher no-show rate. However, the model recall gap was much larger than the underlying outcome-rate gap. This showed that the model relies heavily on booking lead time and under-detects short-lead no-shows.

This finding supported the threshold retest. Short-lead recall improved from 23.0% to 37.2% at threshold 0.45, but the limitation remained.

Cross-track validation Data Science test -> Data Analytics validation -> threshold action -> retest -> documented limitation.

10. Model suitability

The tuned Gradient Boosting model is suitable as a moderate-risk screening and decision-support tool for HealthConnect. It should not be used as a standalone decision system.

Its 0.72 ROC-AUC shows useful predictive signal. It may help prioritize appointments for low-risk administrative support, such as additional reminders. Predictions should still be interpreted with caution because short-lead and Specialist Consultation no-shows are harder to identify.

11. Limitations and risks

- The dataset is fictional and does not establish real-world clinical performance.
- Short-lead no-shows remain difficult to identify.
- Specialist Consultations showed weaker recall than other appointment types.
- Threshold changes improve recall at the cost of more false positives.
- Observed reminder relationships are associations, not causal effects.

12. Week 8 recommendations

- Use the tuned Gradient Boosting model as the final Data Science candidate.
- Validate the saved model and preprocessing requirements during final integration.
- Keep the 0.50 threshold as the default and document 0.45 as a higher-recall option.
- Highlight short-lead appointments as a known limitation.
- Present model performance and limitations clearly in the final HealthConnect solution.

Conclusion

Week 7 moved the model from simple comparison to systematic testing. Testing found moderate overfitting, tuning improved held-out AUC from 0.68 to 0.72, and segment analysis revealed where the model still struggles. The final candidate is better validated and ready for Week 8 integration.